GUNJAN KUMAR MISHRA

Distributed Acoustic Sensing | Scientific Machine Learning | Computer Vision

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RESEARCH PROFILE

Research Engineer in Computer Vision with independent research in distributed acoustic sensing (DAS), landslide-alert evaluation, submarine sensing, and hydrological forecasting. Research interests center on leakage-aware evaluation, physics-aware representation learning, and reliable transfer across sites and sensing conditions. Long-term aim: integrate geophysical sensing and visual observations for natural-hazard early warning and infrastructure monitoring in Nepal.

DAS RESEARCH & MANUSCRIPTS

Independent research | Ongoing

Lead literature review, dataset acquisition, data engineering, experimental design, statistical evaluation, reproducibility auditing, and manuscript development for sensor-data research.

A Leakage-Aware Retrospective Benchmark for DAS-Based Landslide Alerts and Cross-Site Representation Transfer

Submitted manuscript

Focus: retrospective landslide-alert benchmarking and transfer across monitoring sites. Related slope-monitoring work audits chronology confounding, event-level evaluation, and limits on early-warning claims using Hollin Hill and Brienz datasets.

Physics-Paired Contrastive Learning Reveals Direction-Dependent Transfer in Submarine DAS Phase Tracking

Manuscript under refinement

Focus: physics-paired representation learning and direction-dependent transfer in submarine DAS phase tracking.

Structure-Preserving Short-Horizon Prediction of Submarine DAS Coherency at Observed Separations

Manuscript under refinement

Focus: short-horizon prediction of DAS coherency while preserving signal structure and restricting interpretation to observed spatial separations.

HYDROLOGY RESEARCH

Research Assistant | Pulchowk Campus, Tribhuvan University

Apr 2024 - Oct 2024

- Compared Temporal Fusion Transformer, LSTM, Random Forest, and process-based forecasting models using 40 years of daily hydrometeorological data across eight stations.
- Defined temporal splits, antecedent variables, and seasonal error analyses; documented underprediction of monsoon extremes and model limitations in an EarthArXiv preprint.

Attention-Based Deep Learning for Runoff Forecasting | EarthArXiv preprint

EDUCATION

B.E., Electronics, Communication and Information Engineering

Tribhuvan University, Pashchimanchal Campus, Pokhara, Nepal | 2018 - 2024

Full scholarship awarded through entrance-examination merit. Final project: real-time ball detection and trajectory prediction using YOLOv9 and Kalman filtering.

COMPUTER VISION & RESEARCH ENGINEERING

Research Engineer | Angelswing Inc.

Dec 2025 - Present | Remote, South Korea

- Developed and benchmarked YOLO, DETR, RF-DETR, and SAM pipelines on drone and construction-site imagery; evaluated accuracy, latency, model size, robustness, and failure modes.
- Integrated DETR detection with SAM segmentation to produce object-level masks for spatial and site analysis.
- Built FastAPI/Uvicorn and DepthAI services for real-time OAK-D and Arducam streams, exposing inference results, processed frames, depth, and spatial outputs.
- Prototyped image-to-3D raycasting, geolocation, drone-coverage analysis, time-lapse stabilization, rangefinding, and photogrammetry-versus-LiDAR workflows; managed datasets and model artifacts on AWS EC2/S3.

SELECTED TECHNICAL PROJECTS

3D Spatial Mapping & Image-to-World Geolocation

- Developed camera/LiDAR workflows mapping detections and selected image pixels into 3D site models; generated spatial coordinates and geolocation attributes for operational analysis.
- Built drone-coverage heatmaps and traceable processing workflows to identify over-scanned and under-scanned regions from source frames and 3D outputs.

Language & Vision-Language Systems

- Built text/image workflows with schema-validated outputs, tool calling, retrieval, fallback handling, and FastAPI services; evaluated open-source models with vLLM for quality, latency, throughput, and resource use.

TEACHING & MENTORSHIP

DSML Trainer (Part-time) | Logispark Technologies Pvt. Ltd.

Sep 2025 - Present

Design and deliver project-based training in Python, data science, machine learning, model evaluation, and deployment.

Assistant Lecturer & Final-Year Project Reader | Herald College Kathmandu

Oct 2024 - Nov 2025

Taught advanced Python, high-performance computing, and big-data analytics; evaluated final-year projects from literature review and methodology through implementation and documentation.

RESEARCH METHODS & TECHNICAL SKILLS

Sensing & modelling: DAS time series; signal processing; hydrometeorological forecasting; camera geometry; drone imagery; camera/LiDAR workflows.

Learning & vision: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, YOLO, DETR, RF-DETR, SAM, OpenCV, DepthAI; LLM/VLM workflows.

Evaluation: Temporal splits, event-level analysis, cross-site transfer, benchmarking, ablation studies, error and uncertainty analysis, reproducibility auditing.

Engineering: Python, SQL/PostgreSQL, Git, Linux, FastAPI, Uvicorn, AWS EC2/S3, vLLM, Streamlit.

ADDITIONAL MANUSCRIPTS & TRAINING

GLoW YOLOv9 and DETR for Enhanced Billboard Detection | Manuscript

Typing with Emotions | TechRxiv preprint

Selected training: TensorFlow: Advanced Techniques; Machine Learning Specialization; Microsoft Certified: Azure AI Fundamentals.

IELTS Academic: Overall Band 8.0